



Visual Analysis Transcript

Welcome to Bitsplanet Digital. Today we will be continuing our learning journey for the course Data Visualisation Storytelling. We'll be now looking at visual analysis in the Tableau environment and then take it from there to build your dashboards.

We have seen till now how to connect with data, explore joints and plot some simple graphs. We will now understand how to carry out a visual analysis for the data set that you have and try to find insights trends and from there try to build a data story. Remember the purpose of data visualisation is to convey a consistent story to your audience so that they can form their own insights and also enable some amount of interactivity so that you have control on the both author narrative and the audience narrative, balancing the views that you want to present and also enable the audience to derive their own insights.

Now let us look at the various steps of visual analysis. This visual analysis has typically the task that is assigned to you. For example, you want to construct an executive dashboard or you want to present the underlying data trends, that's the task and for that you would get relevant data and in our case we are choosing this particular sample used superstore data so that you can also explore the examples that are readily given in the Tableau.

It's a very standard Excel that is a standard data set that is used. So I'm just going to connect from the beginning so that it is easy for you. So all of you can okay all of you can log in to your Tableau workspace and then connect this from the data source.

So I'm using the EU sample store data sample superstore data which we have explored earlier also. Now what are the steps that are there in the visual analysis? You have a task, you are getting data, then we have to choose the visual mapping, view the data, develop insight and act and share. So we have to start with questions about the basic data that we have and for that we need to understand the data structure first and then choose an appropriate visual and how is it represented.

Once we have the data we'll be able to develop insights and then share the findings. So in Tableau environment how does it work? It works by connecting the sheets. So you are working on a one particular visual in a sheet and then for other visuals you develop another sheets and finally together you stitch them as a dashboard.

Before that let us try to connect data. Okay so we are just dragging the orders and we know these are the fields here row ID, order ID and we also have the people's Excel sheet we can so there is no direct connection here then we have the returns data sheet where the order ID has been identified as the relationship. Now looking at people okay so what do people have field name physical table let us update okay person and region that is only region that is there.



So in orders do we have a region? Yes we have a region in orders. So we have to develop a particular relationship here between people and orders if you want to explore them. Okay so select a relationship this is two returns relate to people I'm trying to relate to people okay automatically region is also recognised.

So now we are ready to explore the data we have established the basic connections between the sheets and it is going to be through particular identifiers order ID in case of orders and return sheet and it is region in terms of order and people. Now let us quickly go to the sheet and try to visually analyse the data. So you have people sheet return sheet and the order sheet.

Now first let us look at the different types of orders that are there. So you have the product name let me get the product name first we'll understand what is there in the product name. So there are so many products which we cannot possibly understand we need to find a higher hierarchy so where I'm just trying to get the segment.

So now you understand what is better is used to the segment consumer segment corporate segment and home segment these are the three segments that we have here. So what we are going to do we are going to understand our analysis with this segment then we have the with this segment we also have the sales okay. So so these are the sales okay so this is the segment and this is the sales.

So the first primary thing this is the business data which is capturing the sales on different segments across various stores or regions for this particular company. So this is the first insight that we got so this is primarily trying to understand sorry we are trying primarily trying to understand what is contained in the data what is the basic task. So the basic task is to analyse sales so this is the fundamental question you want to analyse sales and there are so many other layer variables that can be brought into to understand the relationship between sales growth across segments okay.

So this is what we can try to do okay we can also put one more element here where we have this is geographical variable we have the state so if you can put state before segment so you'll have the state by sales right and you can also use filters right you can also use filter to segment certain data or exam so you can also use the other variables like you have made segment but you can move segment to filter okay and I don't want to see all I just want to understand only the consumer sales so I'm just apply okay I'm just using the consumer sales so this is customer segment filter I have added so I can understand this is the sales is happening and it is in highest in California now what we can also do is to try to sort this data and just sorting this data so I can clearly see which state has highest sales in the consumer segment okay so this is something I can do simultaneously right and then we can try to understand which states are having highest sales and which are having very lowest sales in this particular category now the other way to do this is to take a stacked bar okay and then remove the filter



so we can remove the filter okay so you are having state in sale then I am getting segment also as the column okay this is a sorry segment as zero so this is a very interesting way to see that you have a state where there is highest number of sales in across segments it is ordered across all the three major segments that you have so what are the insights that you are having so just looking at the consumer sales a couple of conclusions that we can have we can draw is of the total sales majority of sales are coming from consumer segment followed by corporate and then the home office and a California and New York are the major states where the sales are higher in absolute levels then you have the you have a Texas Washington Pennsylvania these are the other states where there are higher sales and these are smaller states where there is lesser sales so now these are certain insights that we can draw immediately by looking at a simple graph like a stacked bar chart arranged as per segment and analysing the sales now what you can also try to understand is whether this let us try to duplicate this sheet so once we have duplicated this I am trying to okay understand this I am just removing this amount so let us try to see the returns so whether okay so maybe yeah returns I should have done this as a count so there is a numerical measure I should not use the categorical variable returns count so and this using the returns count so you can also see there are fewer returns in other states but California expectedly where the larger number of sales are there returns are also higher and relatively you have lesser returns in other states but some states do have certain amount of returns so this is a simple conclusion that you can take but at the point that I wanted to demonstrate was you are trying to connect variables from one sheet with variables in the other sheet and draw a combined inside this is happening because in the background tableau is doing the relationship function in the data source itself where it is trying to establish where it has already established the connections between these two sheets and once you hover on the noodle you will get what is the relationship that is there and how is it affecting your data in this case orders to people is connected via region then you have the orders and returns which are connected through the order ID now what interesting thing that you can also do is you can also bring in okay people here so let me also put okay maybe I can use it as filter okay so I can just see region I am looking only ask the East region so I have the East region sales and the returns or else I can remove the East because state becomes too large a variable and remove I am taking state to the filters okay taking state to the filters then I am putting region as the bifurcation criteria so you can see which region and what are the returns so for example West region which I has California okay so you have the highest number of returns and the way that it the particular sheet though people and returns do not have anything in common they are getting connected through the order sheet so this is something we can immediately do once we have a very clean relationship between the sheets generally if you are dealing in a business environment it is ideal to have one connector which is common across all the sheets or even if you don't have in that case you can always try to use a sequential connectors and tableau will connect all of them as if there are in a single a data set and allow you to do the transformations so this is a very simple visual analysis that we are trying to do here so let us also look at couple of other things that we can do okay so what I am trying to do is like I have the

region so I am putting region here okay so then I have the subcategory okay I have put the subcategory and then I'm having the sales so this is one subcategory by sales then what I'm doing I'm trying to okay I just felt that okay this is not right way to present it I want to shift between columns and rows I can click this button so this is what it will look like and I am just sorting it so you can always understand it's easy to sort categories okay and it is easy to sort categories in the logical order that is there in the data so that you can convince your audience or there is a pattern which is cognitively pleasing to them so one element that is missing here is we are trying to understand sales as per subcategory but again in the subcategory I am not able to clearly see which is the main segment whether it is home office consumer or corporate so you can just get the category here in front of subcategory okay so now we have this very clean graph so you can extend that and arrange on or try to on it so this is what we have done so we can also make adjustments to colour okay so so we can make adjustments to colour so this is something we can do we can also bring the subcategory to colour yeah so this is how we can do different colours and then we can select on a particular element then we can understand how is it behaving across regions so this is very simple visual analytic tools that will help you to understand what is happening in your data and derive insights for your customers okay so this is something very intuitive and easy for you to understand let us say first you want to compare the particular two categories you want to try our talk about binders we have talking only about binders across all regions then you have the values that are there and maybe we can also put labels okay so mark labels so we have the values also here so very clearly you can see almost all the sales of binders is similar across all regions whereas if you take machines the there is very high difference between different regions so like this you can do a visual comparison and drawing sites so before you publish your dashboard it is important that you understand what are the various dimensions in your data how they are performing on top so now let us also quickly try to do a map visual okay so I'm just trying to get a map okay so I just okay I don't need I need a field map I don't need a dot map it's also important to choose the right map for your visualisation so I am just using the city in rows okay I'm just interested in city in rows and I just want to understand not sorry sales okay it's the map visualisation so we are having latitude longitude which has been generated then I am adding sales you can next try a presenting the data on a map you can simply have to drag for example you can open a new sheet so what am I doing I'm bringing clicking on the state variable which is a location variable tableau will automatically understand this is a geographical data and presents the map locations as dots okay and then you can try to bring a numerical variable let us say sales onto the colour okay so you can just reorder so you have the colours for different states and the sales are indicated as a palette here so you can then format or edit colours you can choose you want automatic or you want for example orange blue diverging okay so this is how it will look like so you can try to understand this so now all these are ready for you in the sense you have different colours of I mean different visuals that you have done so you can use the which is more colour friendly so let us yeah this looks more colour friendly in terms of trying to understand you can clearly understand more major sales are



coming from California and New York and rest of other sales are very comparatively little lesser so this is how you can see and Texas is the next big centre of course Washington all these are related to the population and the type of companies that are there so once you have done this maybe I can also add okay I have sales and then yeah what I want to do I want to add more elements into this we can try to do that I can I can just pull region so you can see now what actually you don't know which regions represent what states you can simply do this so how is this useful you can start writing or explaining to the people the eastern region or the central region so this is central which has Texas so this is the thing then you have New York in the East region so South region has no large state where sales are coming from then you have California in the West region which is a major contributor so if you were to see on an aggregated basis you have in each of the regions except for Central South region in each of the regions it is one major state or one major player in the game so you can compare to you can compare so you can compare these things individually and try to make insights for your particular graphic so this is the way that you will try to use different visual cues that are there which are tools that are there in your tableau space to derive insights again what I would advise you is to explore the data set with different visuals try and test different elements that are there especially the sorting function that you have in the tableau which helps you to our there is one more point here always remember to connect your data sheets so that you can also achieve the cross functionality as we have just achieved in this particular exercise therefore you are able to draw your insights consistently across sheets and present a unified view to your audience we will now use this element for building your dashboards thank you.

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